PROJECT REPORT

On

AN AI – POWERED CROP DISEASE DETECTION AND AGROCHEMICAL RECOMMENDATION FRAMEWORK

A Project Report submitted to

Alliance University, Bengaluru

For the award of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

by

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CERTIFICATE

This is to certify that the project work entitled **"An AI-Powered Crop Disease Detection and Agrochemical Recommendation Framework"** submitted by Chalini M(2023BCSE07AED440), Spandana Basava Kulal (2023BCSE07AED441) in partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science and Engineering of Alliance University, is a Bonafide accomplished under our supervision and guidance during the academic year 2026-2027. This thesis report embodies the result of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

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DECLARATION

We hereby declare that the project entitled **"An AI-Powered Crop Disease Detection and Agrochemical Recommendation Framework"** submitted by me/us in the partialfulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of ASAC, Alliance University, is a record of my/our work carried under the supervision and guidance of Dr. Qwasar Gulzar- Assistant Professor

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

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ABSTRACT

The agricultural sector maintains its vital position because it produces food while enabling farmers to earn money through their farming activities. The failure to detect crop disease during their initial stages leads to plant development problems which produce lower crop yields and financial damage to farmers. Farmers need to identify disease through visual inspections and expert advice to determine which treatment methods will work best. The process requires long hours to complete which results in wrong chemical applications because farmers make wrong disease identification choices. The project introduces an AI-based system which identifies crop diseases through images and recommends specific agrochemical treatments for each detected disease.

The framework uses deep learning techniques to identify crop disease from leaf images. A labelled image dataset, such as PlantVillage, can be used to train or fine tune – a suitable convolutional neural network (CNN). The system performs image resizing and normalization during pre-processing to transform input images into the necessary structure which the model requires. The system connects to a recommendation module which provides suitable agrochemical and crop-management guidance based on agricultural data that has been verified. The system delivers its result through a simple interface which enables farmers to understand both the diagnosed condition and treatment alternatives they can select.

The proposed framework seeks to establish a method which detects disease during their initial stages while it provides farmers with useful crop-management information and helps them select proper agrochemical treatments. The system functions as a decision-support tool for farmers and agricultural practitioners because it combines disease detection with treatment guidance within a single application. The framework will receive its evaluation through classification metrics which consist of accuracy and precision and recall and F1-score and the recommendation module will undergo validation.

LIST OF FIGURES

Figure No	Description	Page No.
3.1	System Architecture Diagram	4
4.2.2	Flowchart	11
5.11	Dashboard Screenshot	16
5.1.2	Output Result Graph	18

LIST OF TABLES

Table No	Description	Page No.
4.1.1	Software Tools	7
4.1.2	Hardware Requirements	8
4.1.3	Frameworks / Libraries	19
5.1.3	Output	4
5.2.2	Evaluation Results	21
5.3.1	Comparison with Existing Systems	22-23

LIST OF ABBREVATIONS & SYMBOLS

Abbreviation / Symbol	Meaning
CNN	Convolution Neural Network
API	Application Programming Interface
UML	Unified Modeling Langauge
DL	Deep Learning
ML	Machine Learning
AI	Artificial Intelligence
OpenCV	Open Source Computer Vision Library
XAI	Explainable Artificial Intelligence
X	Preprocessed image tensor
C	Total number of classification categories (38)
KB	Agricultural knowledge base